DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION

Course Title: Mathematical Analysis for Computer Science
Course Code: CSE-3201
3rd Year 2nd Semester
Session: 2017-18

Time: 3 Hours

[Answer Any FIVE Questions]

Marks: 60

1. Mathematical analysis is the key to opportunity. No longer just the language of science, now it contributes in direct and fundamental ways to business, finance, health, and defense. For students, it opens doors to careers. For citizens, it enables informed decisions. For nations, it provides knowledge to compete in a technological community. Now your task is to,

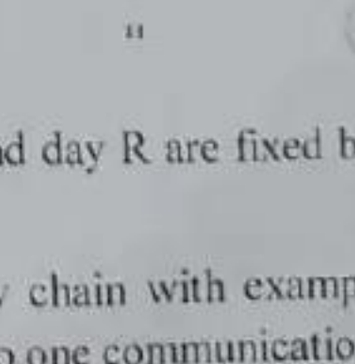
a) How mathematical analysis is fundamental to computer science and engineering? Justify your answer. [3]

b) List the first three terms for the following series from a1 to a4. [3]

- $a_n \quad \sum_{i=1}^{n}(2i - 1)$
- $a_n \quad \sum_{i=1}^{n}(3i^2 - 3i + 1)$
- $a_n \quad \sum_{i=1}^{n}\sum_{j=1}^{n} 1$
- $a_n \quad \sum_{i=1}^{n}\sum_{j=1}^{n} i$
- $a_n \quad \sum_{i=1}^{n}\sum_{j=1}^{n} j$

c) What do you mean by conditional probability and expectation? A box contains 4 black chips and 2 blue chips. If two chips are selected at random without replacement, what is the probability that the chips are different colors? [3]

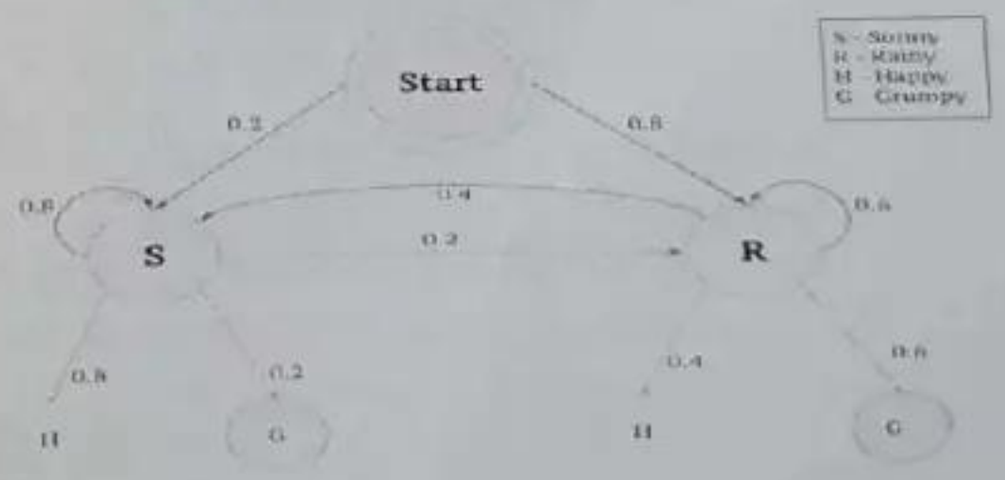
d) A "Venn diagram" with three overlapping circles is often used to illustrate the eight possible subsets associated with three given sets. Can the sixteen possibilities that arise with four given sets be illustrated by four overlapping circles? Justify your answer. [3]



2. a) Write down the differences among dynamic programming, markov chain and monte carlo method with examples. [3]

b) What do you mean by hidden markov model? Write down the following questions. [6]

i) Draw the transition matrix with the probability if emission of 3 days are G, H, G given that S, R and S respectively.



ii) What will be the probability if emission of 3 days are H, G, H considering first day S and second day R are fixed but third day is random. What is the most possible outcome for the event.

c) Write short notes on transient, recurrent and classes in Marcov chain with examples. "A Markov chain is called irreducible if and only if all states belong to one communication class" justify your answer. [3]

3. a) Probability is the measure of the likelihood that an event will occur. Probability quantifies as a number between 0 and 1, where, loosely speaking, 0 indicates impossibility and 1 indicates certainty. [4]

i. A die is rolled, find the probability that the number obtained is greater than 4.
ii. Two coins are tossed, find the probability that one head only is obtained.
iii. Two dice are rolled, find the probability that the sum is equal to 5.
iv. A card is drawn at random from a deck of cards. Find the probability of getting the

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